SIPs,[1604] and self-aggregators will be able to generate consolidated market data for their own use, and the use of their broker-dealer and registered investment advisor affiliates.

The Commission understands that there is an inherent conflict of interest in that the exchanges, as voting members of the Equity Data Plan Operating Committees, may not be incentivized to improve the content or latency of SIP data.[1605] For example, certain exchanges have developed proprietary data products with reduced latency and expanded content (i.e., proprietary DOB products), while not taking similar action on these committees to enhance the data products offered by the Equity Data Plans.[1606] These proprietary DOB products have evolved to be considered competitive necessities by many market participants and are offered at premiums to exclusive SIP products.[1607] Similarly, some exchanges have developed limited TOB data products, offering them at a discount compared to the SIP data, while the exclusive SIPs have not developed less expensive SIP products.[1608] The exchanges have continued to

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1604 See supra Section III.

1605 A number of commenters agreed that the SROs have a conflict of interest. See, e.g., Wellington Letter at 1; IEX Letter at 2; Fidelity Letter at 3. See also Proposing Release, 85 FR at Section III.A. and n.267 (describing an exchange-led initiative to enhance the SIPs). While the new Equity Data Plan, required to be filed pursuant to the Governance Order, is required to be designed to address these conflicts of interest, it would not eliminate them.

1606 See Governance Order, supra note 1128 at Section II.B.1. Commenters agreed that the improvements to the SIPs have not kept pace with the improvements in proprietary feeds. See, e.g., State Street Letter at 2 ("Over time, improvements have been made to the SIPs, but those improvements have not kept pace with the alternative data feeds that the industry can and is often required to access"); Wellington Letter at 1.

1607 See id.

1608 See id.; see also Proposing Release, 85 FR at n.25. The Commission did not receive comments disagreeing with this characterization of the relationship between the exclusive SIPs and TOB feeds.

develop and enhance their proprietary market data businesses—which generate revenue that, unlike SIP data revenues, do not have to be shared with the other SROs—while remaining responsible for the governance and operation of the Equity Data Plans, including content, infrastructure, and pricing, as well as data consolidation and dissemination.[1609]

The Commission believes that there are two additional factors related to the Equity Data Plan processors that may impede improvements to the dissemination of SIP data. First, pursuant to Regulation NMS, each exclusive SIP has exclusive rights to collect trade and quotation data related to NMS stocks from multiple SROs and then aggregate and disseminate market data to market participants.[1610] This structure may further impede improvements in the dissemination of SIP data[1611] because Equity Data Plan participants that govern exclusive SIPs do not have incentives to innovate due to the lack of competition in dissemination of SIP data.

Second, the exclusive SIPs are either SROs themselves or affiliates of SROs.[1612] This gives the SROs a dual role in that they serve as both existing plan processors and as entities selling directly their own proprietary market data products that can reach market participants faster than SIP data, or as affiliates of entities that do so. As discussed in the Proposing Release, this may create an additional conflict of interest that could provide incentives making the Equity Data Plan participants that oversee the Equity Data Plans reluctant to improve the content and latency of the SIP data, because a divergence in the usefulness of SIP data provided by the

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1609 See Governance Order, supra note 1128, at Section II.B.1.

1610 See Proposing Release, 85 FR at n.21 and accompanying text.

1611 See infra Section V.B.2(b).

1612 See Proposing Release, 85 FR at n.42.

exclusive SIPs as compared to the proprietary data feeds increases the value of the proprietary market data products.[1613]

The Commission is concerned that Regulation NMS and the Equity Data Plans have not kept pace with the needs of market participants as markets, trading systems, and technologies have changed dramatically. While the exchanges have developed individual proprietary data products to meet the needs of some market participants, the Commission believes that there should be improvement to, and modernization of, the national market system to fulfill the goals of Section 11A of the Exchange Act and to meet the current core data needs of all market participants. The Commission is concerned that the lack of modernization to the content and dissemination of SIP data compared to proprietary data feeds has contributed to the development of a two-tiered system in which certain market participants who are able to afford, and choose to pay for, the exchanges' relatively more expensive proprietary DOB data feeds and associated connectivity and transmission offerings receive more content-rich data faster than those who do not receive these data feeds.[1614]

Some market participants are unable to rely solely on SIP data to trade competitively and execute investor orders in today's markets.[1615] SIP data currently does not include some

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1613 Commenters agreed with this assessment. See, e.g., BestEx Research Letter at 1 ("SIP operators have little incentive to provide better content at more competitive prices with lower latency because it may cannibalize their own direct feed business") and 4 ("The bigger the differences in content between direct data feeds and SIP, the more power exchanges have in setting their own prices for market data.").

1614 See supra Sections I.A and I.B.

1615 See supra Section II.C.2(c); see also id.; Proposing Release, 85 FR at Section III.C.1(a); infra Section V.B.2(c). A number of commenters agreed that market participants may not be able to rely on the SIP to trade competitively. See, e.g., DOJ Letter at 2 ("[P]articipants that rely solely on SIP Data could be at a competitive disadvantage to

important data elements such as odd-lot quotations (except to the extent that odd-lot quotations are aggregated into round lots pursuant to exchange rules),[1616] depth of book data, and information about orders participating in auctions.[1617] Moreover, there is a substantial latency differential between market data provided via the exclusive SIPs and proprietary data products delivered by the exchanges directly to market participants or to market data aggregators as part of exchange proprietary data feeds.[1618] The latency and content disparity between SIP data feeds and proprietary DOB data products has the effect of increasing the market participants' demand for proprietary products to the extent that some brokers-dealers stated they view acquiring such products as a competitive necessity.[1619] Additionally, market participants have stated that the higher prices charged for some exchange proprietary DOB feeds and associated connectivity and transmission limits the number of broker-dealers accessing these feeds and places those that do

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those that rely on multiple sources of market information, including Prop Data"); MFA Letter at 2; BlackRock Letter at 2; Wellington Letter at 1; IntelligentCross Letter at 4.

1616 See supra Section II.C.2(b); see also Proposing Release, 85 FR at Section III.C.1(a).

1617 Only limited auction-related information is currently included in SIP data. See supra Section II.G; see also Proposing Release, 85 FR at Section III.C.3(a).

1618 See infra Section V.B.2(b).

1619 A number of commenters agreed that broker-dealers need to purchase proprietary DOB feeds in order to trade competitively. See, e.g., Clearpool Letter at 2 ("[B]roker-dealers are compelled to purchase the exchanges' proprietary data feeds both to provide competitive execution services to clients and to meet best execution obligations due to the content of the information contained in proprietary data feeds, as well as the lack of latency in those feeds, both important considerations for brokers"); State Street Letter at 2; Better Markets Letter at 1; T. Rowe Price Letter at 1. Commenters also agreed that there is a disparity between the content and latency of the SIP data feeds and proprietary market data. See, e.g., Committee on Capital Markets Regulation Letter at 2.

not subscribe at a competitive disadvantage relative to other market participants willing and able to spend the money to access these feeds.[1620]

One commenter stated that all of the additional information provided by the proprietary feeds is already available to everyone who needs it.[1621] While the Commission acknowledges that the option to subscribe to proprietary market data is available to all market participants, the Commission is concerned that the national market system needs improvement to fulfill the goals of Section 11A of the Exchange Act and to meet the current core data needs, including data content and latency, of all market participants. The Commission is concerned a two-tiered system has developed in which market participants that do not receive proprietary DOB feeds may be affected in their efforts to seek best execution and otherwise effectively compete with market participants that receive proprietary DOB data feeds. The Commission believes that consolidated market data must reflect all information that is important for a broad cross section of investors and market participants and must do so in a manner that is latency-sensitive.

B. Baseline

The Commission has assessed the likely economic effects of the final amendments, including benefits, costs, and effects on efficiency, competition, and capital formation, against a baseline that consists of the existing regulatory process for collecting, consolidating, and

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1620 See supra notes 26-28 and accompanying text; infra Sections V.B.3(e),V.B.2(f). For example, one commenter stated "[w]e observe increasing concentration in the financial industry – in the asset manager space, the broker/dealer community, and in the liquidity provider/market maker space. There are barriers to entry based on necessary scale to be able to absorb the fixed costs of infrastructure, market data and connectivity," and that "algorithmic executions by broker/dealers cannot in general be competitive if they do not use direct feeds." See NBIM Letter at 3. Additionally, there are indicia that exchanges may not be subject to robust competition with respect to market data. See infra Section V.B.3(b).

1621 See Nasdaq Letter IV at 34.

disseminating market data, and the structure of the markets for SIP data products and for connectivity and trading services.

1. Current Regulatory Process for Equity Data Plans and SIP Data

The current regulatory framework for SIP data relies upon a centralized consolidation model, whereby the SROs provide certain quotation and transaction information for each NMS stock to a single exclusive SIP, which then consolidates this data and makes it available to market participants.[1622] This SIP data includes what historically has commonly been referred to as core data, as well as certain regulatory data related to Commission and SRO rules and NMS plan requirements.[1623]

As discussed in more detail below,[1624] SIP data currently includes transaction information for both round lot and odd-lot-sized transactions as well as quotation information for round lot top of book quotes for each SRO. Additionally, several exchanges, pursuant to their own rules, aggregate odd-lot orders into round lots and report such aggregated odd-lot orders as quotation information to the exclusive SIPs.[1625] Thus, SIP data lacks information on odd-lot quotations at prices better than the best bid and offer and on depth of book quotations (i.e., limit orders resting at exchanges at prices outside of the bid and offer). Additionally, only limited auction-related information is included in SIP data.[1626]

1622 See Proposing Release, 85 FR at Section II.A.

1623 Id.

1624 See infra Section V.B.2(a). See also Proposing Release, 85 FR at Section II.C.1.

1625 See Proposing Release, 85 FR at Section II.A.

1626 See Proposing Release, 85 FR at Section III.C.2.

Currently, the Operating Committees of the Equity Data Plans, which are governed exclusively by the SROs,[1627] select the exclusive SIPs to consolidate and disseminate market data to market participants. The selection process for the exclusive SIPs is organized through a bidding process, and once selected, an exclusive SIP has exclusive rights to consolidate and disseminate market data for a given Equity Data Plan.[1628] Currently, SIAC (a NYSE affiliate) is the exclusive SIP for the CTA and CQ Plans, and Nasdaq Stock Market LLC is the exclusive SIP for the UTP Plan.

Each exclusive SIP is physically located in a different data center.[1629] The exchanges' and FINRA's primary data centers are also located in different locations. Each exchange and FINRA must transmit its quotation and transaction information from its own data center to the appropriate exclusive SIP's data center for consolidation, at which point SIP data is then further transmitted to market data end-users, which are often located in other data centers. The exclusive SIPs do not compete with each other in the collection, consolidation, or dissemination of SIP data.

1627 Under the Governance Order, the Operating Committee of the New Consolidated Data Plan would include non-SRO members. See Governance Order, supra note 1128.

1628 The Nasdaq UTP Plan contains the description of its approach to the selection and evaluation of the processor. See Nasdaq UTP Plan, supra note 10, at 10. The CTA/CQ Plan does not contain a similar provision. See CTA Plan, supra note 10; CQ Plan, supra note 10. Historically, exchanges or exchange affiliates had always been selected to be plan processors.

1629 See Proposing Release, 85 FR at Section II.A. and n.43.

2. Current Process for Collecting, Consolidating, and Disseminating Market Data

In addition to the provision of SIP data pursuant to the Equity Data Plans, the national securities exchanges separately sell their individual proprietary market data products directly to market participants via proprietary data feeds.[1630] Proprietary data feeds may include SIP data elements and a variety of additional data elements and can vary in content from proprietary TOB products to proprietary DOB products.[1631] In addition, in connection with proprietary data feed products, the exchanges offer various connectivity services (e.g., co-location at primary data centers, fiber optic connectivity, wireless connectivity, and point-of-presence connectivity at third-party data centers), which may result in higher speed transmissions.[1632] Typically, proprietary data is transmitted directly from each exchange to the data center of the subscriber, where the subscriber's broker-dealer or vendor (or the subscriber itself) privately consolidates such data with the proprietary data of the other exchanges. This section describes the current content of SIP data and proprietary data feeds, current process of data dissemination, and current process for costs of generating SIP data and proprietary data feeds.

1630 See Proposing Release, 85 FR at Section II.A.

1631 See Proposing Release, 85 FR at Section III.C.2.

1632 See Proposing Release, 85 FR at n.51 and accompanying text; Proposing Release, 85 FR at Section IV.A.

(a) Current Content of SIP Data and Proprietary Data Feeds

As discussed in the Proposing Release,[1633] today SIP data does not include some of the content that certain market participants rely on when handling customer orders and trading.[1634] This difference in content creates significant information asymmetries between market participants who rely solely on SIP data and market participants who also rely on proprietary data feeds.[1635]

A certain portion of market participants do not rely solely on SIP data to trade competitively in today's markets and instead purchase proprietary data from SROs to supplement or even replace SIP data.[1636] In particular, the Commission understands that approximately 50 to 100 firms purchase all of the proprietary DOB feeds from the exchanges and do not rely on the SIP data for their trading.[1637] Conversely, the number of users of the SIP data is much larger (in

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1633 See Proposing Release, 85 FR at Section III.C.

1634 A number of commenters agreed with this statement. See, e.g., Clearpool Letter at 11; IEX Letter at 5–6; Virtu Letter at 2; DOJ Letter at 2. See also supra notes 1615, 1619 and accompanying text.

1635 Commenters agreed with this assessment. See, e.g., MEMX Letter at 2 (acknowledging that "information asymmetries exist between market participants consuming consolidated data disseminated through the" exclusive SIP feeds and "market participants consuming proprietary data feeds directly from national securities exchanges"); Clearpool Letter at 15; Schwab Letter at 3.

1636 See Proposing Release, 85 FR at Section VI.B.2(a). The Commission believes that when market participants purchase proprietary data feeds to replace SIP data, they also almost always purchase SIP data as a back-up system to proprietary data. See also Proposing Release, 85 FR at n.101.

1637 See Proposing Release, 85 FR at n.140. In addition to using proprietary DOB feeds for non-display purposes, these firms may also use proprietary DOB feeds for display purposes for their employees and clients.

the millions),[1638] suggesting that many users rely on the exclusive SIPs alone. The Commission believes that a large portion of retail investors rely solely on SIP data for trading decisions.[1639] However, some retail investors may use data derived from proprietary feeds from one or more exchanges in order to obtain additional data beyond the NBBO.[1640]

As described in the Proposing Release,[1641] SIP data consists of certain quotation[1642] and transaction data[1643] that the SROs are required to provide to the exclusive SIPs for consolidation and dissemination to the public on the consolidated tapes. Specifically, the SIP data includes: (1)

1638 As of the fourth quarter of 2019, there were approximately 2–3 million non-professional subscribers and approximately 0.3 million professional subscribers across the UTP and CTA/CQ SIPs. Additionally, there were approximately 300 non-display vendor use cases at each of the exclusive SIPs. See, e.g., CTA Plan, Q2 2020 CTA Tape A & B Quarterly Population Metrics, available at https://www.ctaplan.com/publicdocs/ctaplan/CTAPLAN_Population_Metrics_2Q2020.pdf; Nasdaq UTP Plan, Q2 2020 UTP Quarterly Population Metrics, available at https://www.utpplan.com/DOC/UTP_2020_Q2_Stats_with_Processor_Stats.pdf. The Commission understands that there is an overlap in subscribers across the exclusive SIPs.

1639 See Proposing Release, 85 FR at Section VI.B.2(a). Commenters agreed that many retail investors only view core data. See, e.g., Schwab Letter at 2; MEMX Letter at 3. Retail investors may also view proprietary TOB feeds that contain less content than the SIP. See infra note 1651.

1640 One retail broker stated that it "currently offers depth of book products at a reasonable cost for those investors who find the data useful. Providing this data from separate feeds in specific circumstances for investors allows clients to choose what data beyond a national best bid and national best offer ("NBBO") is important and useful to them and avoids overwhelming amounts of information." See TD Ameritrade Letter at 6.

1641 See Proposing Release, 85 FR at Section II.A.

1642 See Rule 602 of Regulation NMS.

1643 See 17 CFR 242.601 (Rule 601) of Regulation NMS.